

- 3 (a) (numerically equal to) quantity of (thermal) energy required to change the state of unit mass of a substance without any change of temperature
(Allow 1 mark for definition of specific latent heat of fusion/vaporisation)
- M1
A1 [2]
- (b) *either* energy supplied = $2400 \times 2 \times 60 = 288000 \text{ J}$ C1
 energy required for evaporation = $106 \times 2260 = 240000 \text{ J}$ C1
 difference = 48000 J
 rate of loss = $48000 / 120 = 400 \text{ W}$ A1
or energy required for evaporation = $106 \times 2260 = 240000 \text{ J}$ (C1)
 power required for evaporation = $240000 / (2 \times 60) = 2000 \text{ W}$ (C1)
 rate of loss = $2400 - 2000 = 400 \text{ W}$ (A1) [3]

4 (a) $c = \left(\frac{1}{2} \right) \rho^2 v$ and $c = 3v/T$

C1